

NORMAL DISTRIBUTION

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Normal Distributions

This is the most common observed distribution of continuous random variables. A **normal distribution** corresponds to bell-shaped curves.

Examples

The following are examples of normally distributed everyday data.

- Grades on a test.
- How many chips are in a small bag of potatoes chips.
- The measurements of distance between two points.
- The heights of students in this class.

Normal Distribution

Why are normal distributions so important?

- Many dependent variables are commonly assumed to be normally distributed in the population
- If a variable is approximately normally distributed we can make inferences about values of that variable

Normal Distributions

- A random variable X with mean μ and standard deviation σ is normally distributed if its probability density function is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(1/2)\left(\frac{x-\mu}{\sigma}\right)^2} \quad -\infty \leq x \leq \infty$$

where $\pi = 3.14159\dots$ and $e = 2.71828\dots$

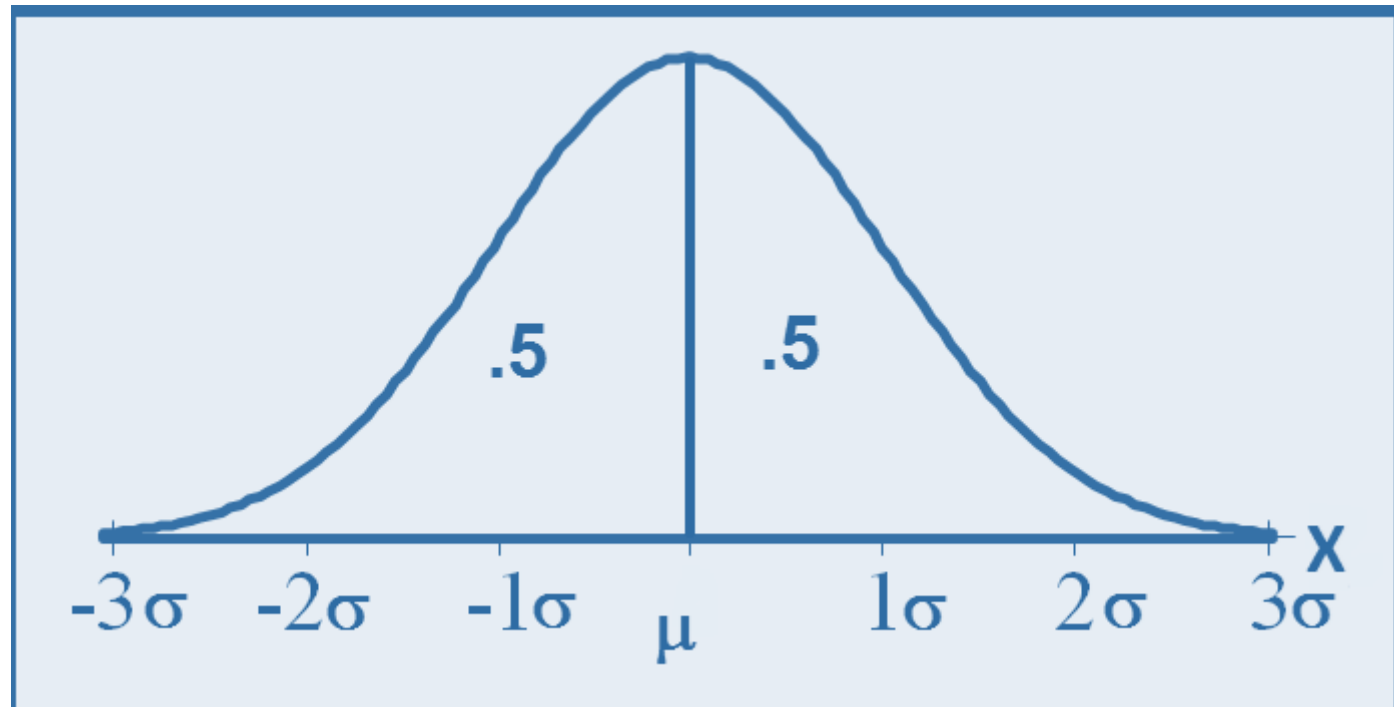
Normal Probability Distributions

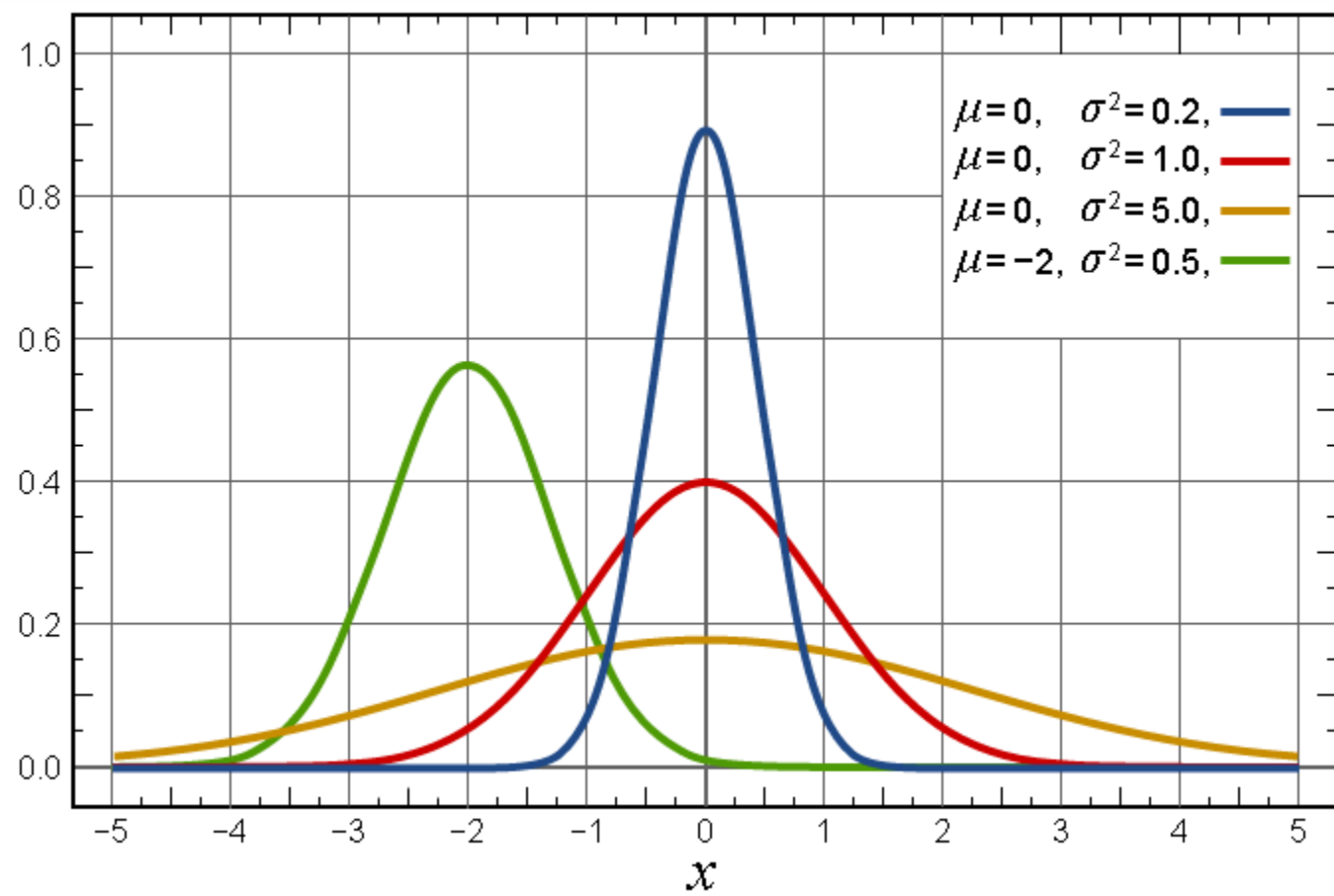
- The expected value (also called the mean) $E(X)$ (or μ) can be any number
- The standard deviation σ can be any nonnegative number
- The total area under every normal curve is 1
- There are infinitely many normal distributions

Characteristics of the Normal Distribution

1. Has a Bell Shape Curve
2. It is symmetrical about the mean μ . The curve on either side of μ is a mirror image of the other side.
3. The mean , the median and the mode are equal.
4. The total area under the curve above x-axis =1

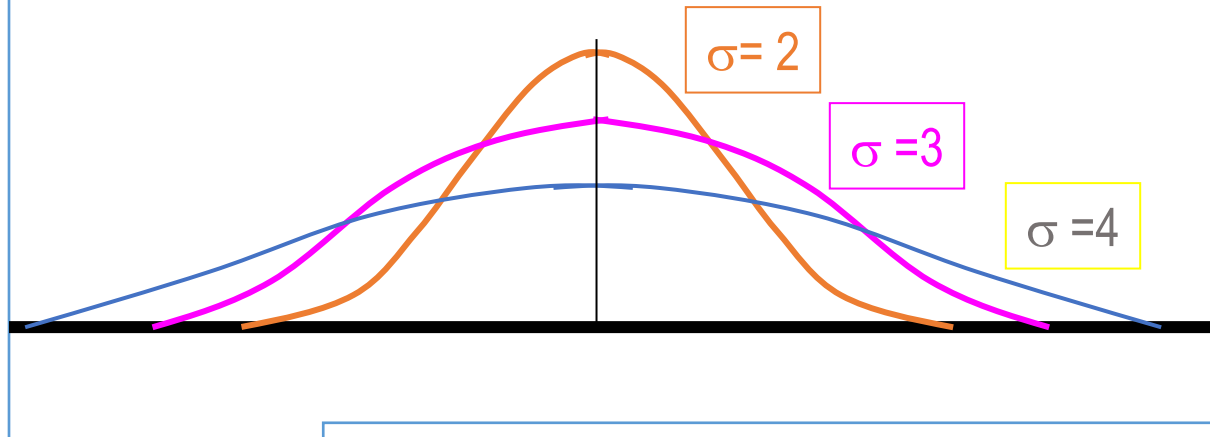
Total area = 1; symmetric around μ



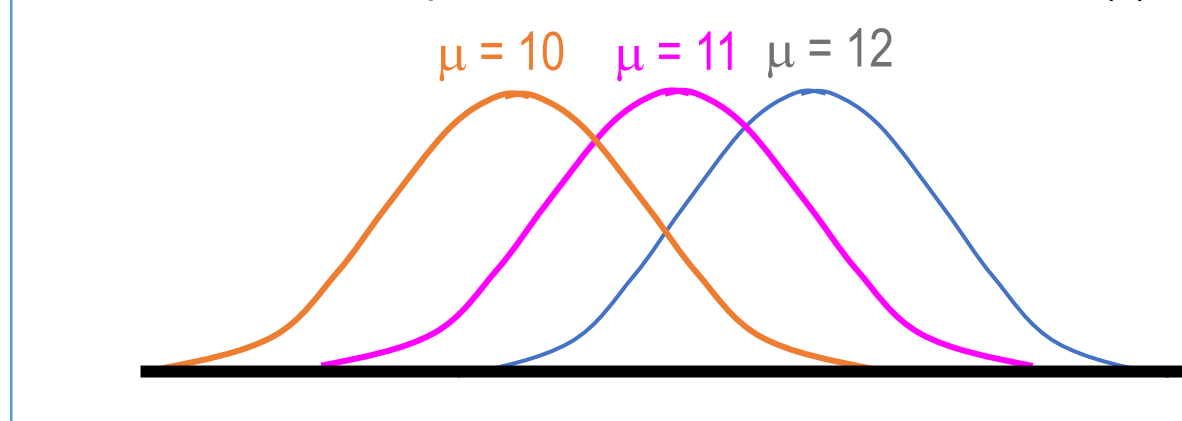


The effects of μ and σ

How does the standard deviation affect the shape of $f(x)$?

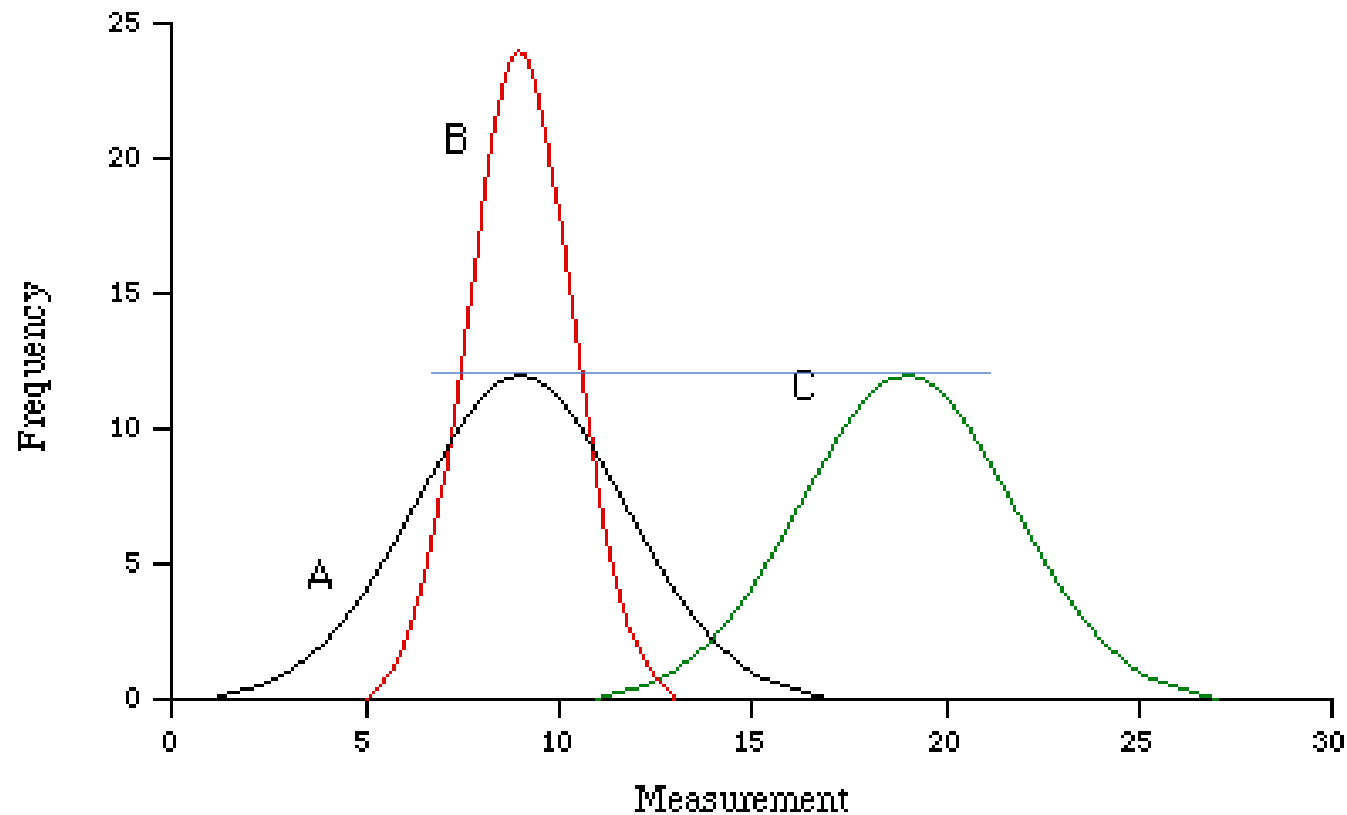


How does the expected value affect the location of $f(x)$?



Effects of μ and σ

changing μ shifts the distribution left or right.
Changing σ increases or decreases the spread.



The Standard Normal Distribution

A normal distribution with a mean of 0 and a standard deviation of 1 is called the standard normal distribution.

Z value: The distance between a selected value, designated X , and the population mean, divided by the population standard deviation,

$$Z = \frac{X - \mu}{\sigma}$$



Z-score

-
- $Z = \frac{x - \mu}{\sigma}$
 - Z indicates how many standard deviations away from the mean the point x lies.
 - Z score is calculated to 2 decimal places.

Z-score

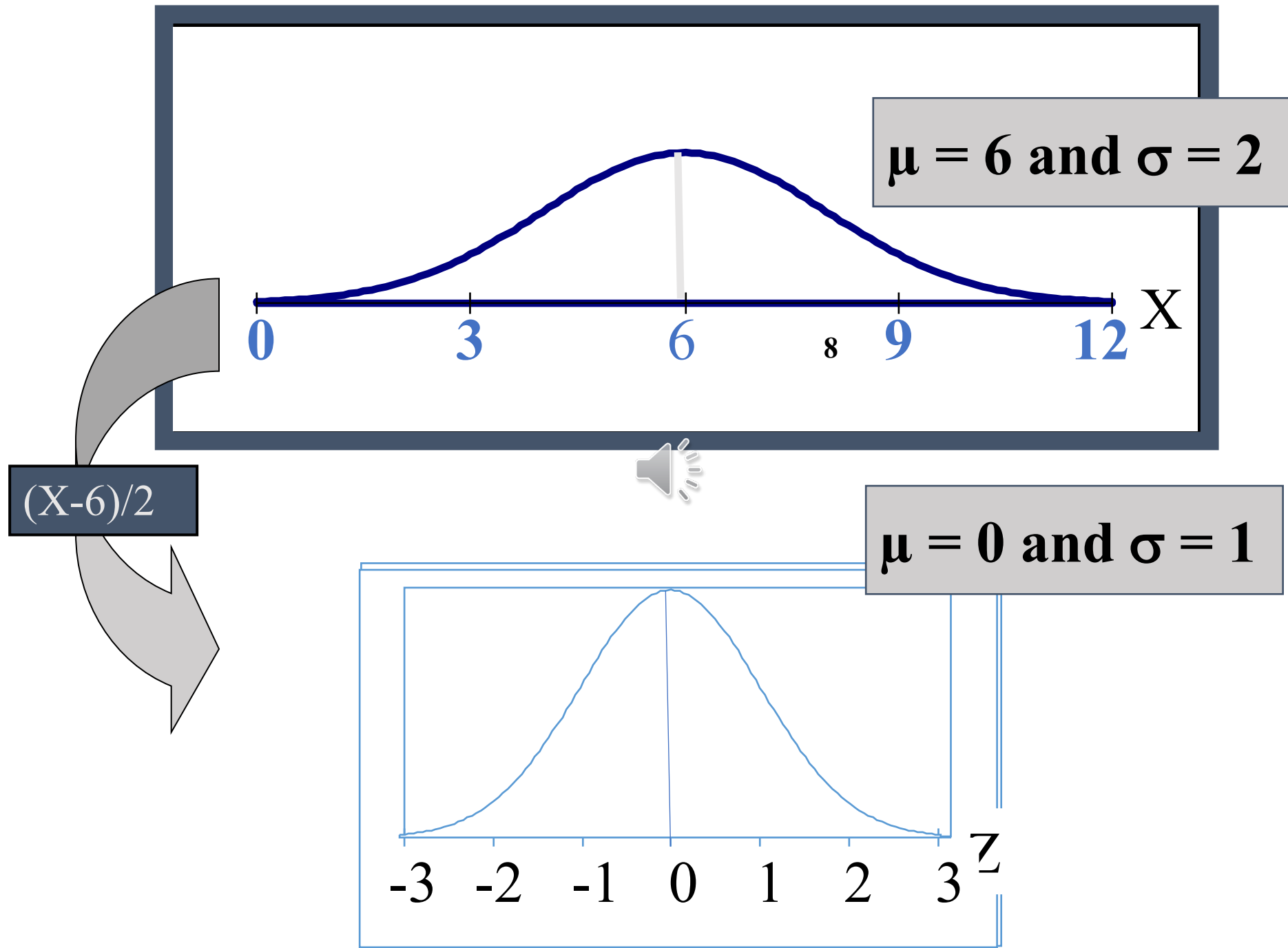
- If X has mean μ and stand. dev. σ , standardizing a particular value of x tells how many standard deviations x is above or below the mean μ .
- Exam 1: $\mu=80$, $\sigma=10$; exam 1 score: 92
Exam 2: $\mu=80$, $\sigma=8$; exam 2 score: 90

Which score is better?

$$z_1 = \frac{92 - 80}{10} = \frac{12}{10} = 1.2$$

$$z_2 = \frac{90 - 80}{8} = \frac{10}{8} = 1.25$$

90 on exam 2 is better than 92 on exam 1



e.g. two tests:

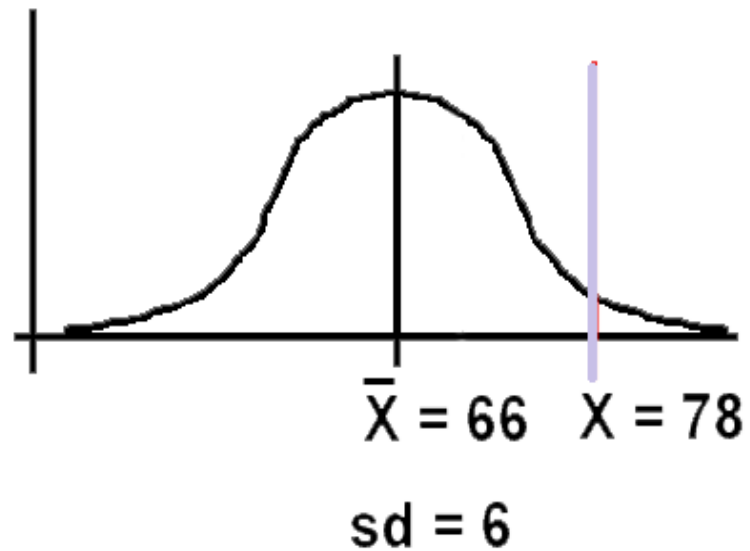
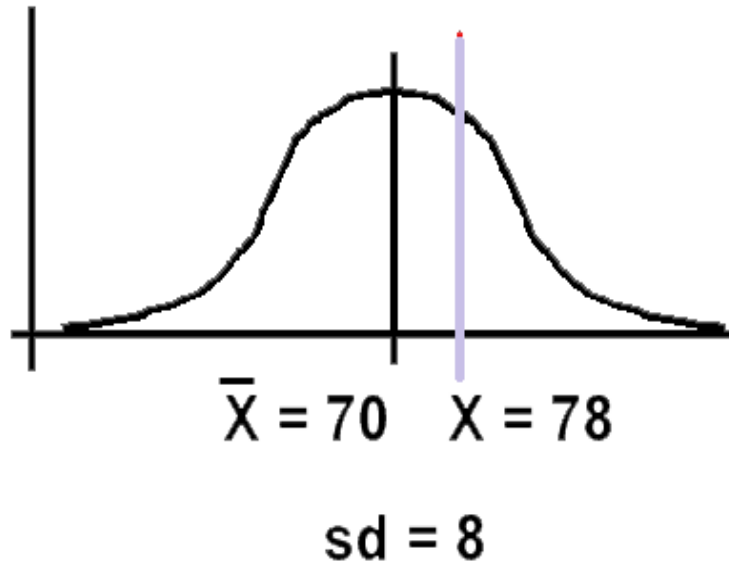
Test A: Fred scores 78. Mean score = 70, SD = 8.

Test B: Fred scores 78. Mean score = 66, SD = 6.

Did Fred do better or worse on the second test?

Test A: as a z-score, $z = (78-70) / 8 = 1.00$

Test B: as a z-score, $z = (78 - 66) / 6 = 2.00$



EXAMPLE

The monthly incomes of recent MBA graduates in a large corporation are normally distributed with a mean of \$2000 and a standard deviation of \$200. What is the Z value for an income of \$2200? And an income of \$1800?

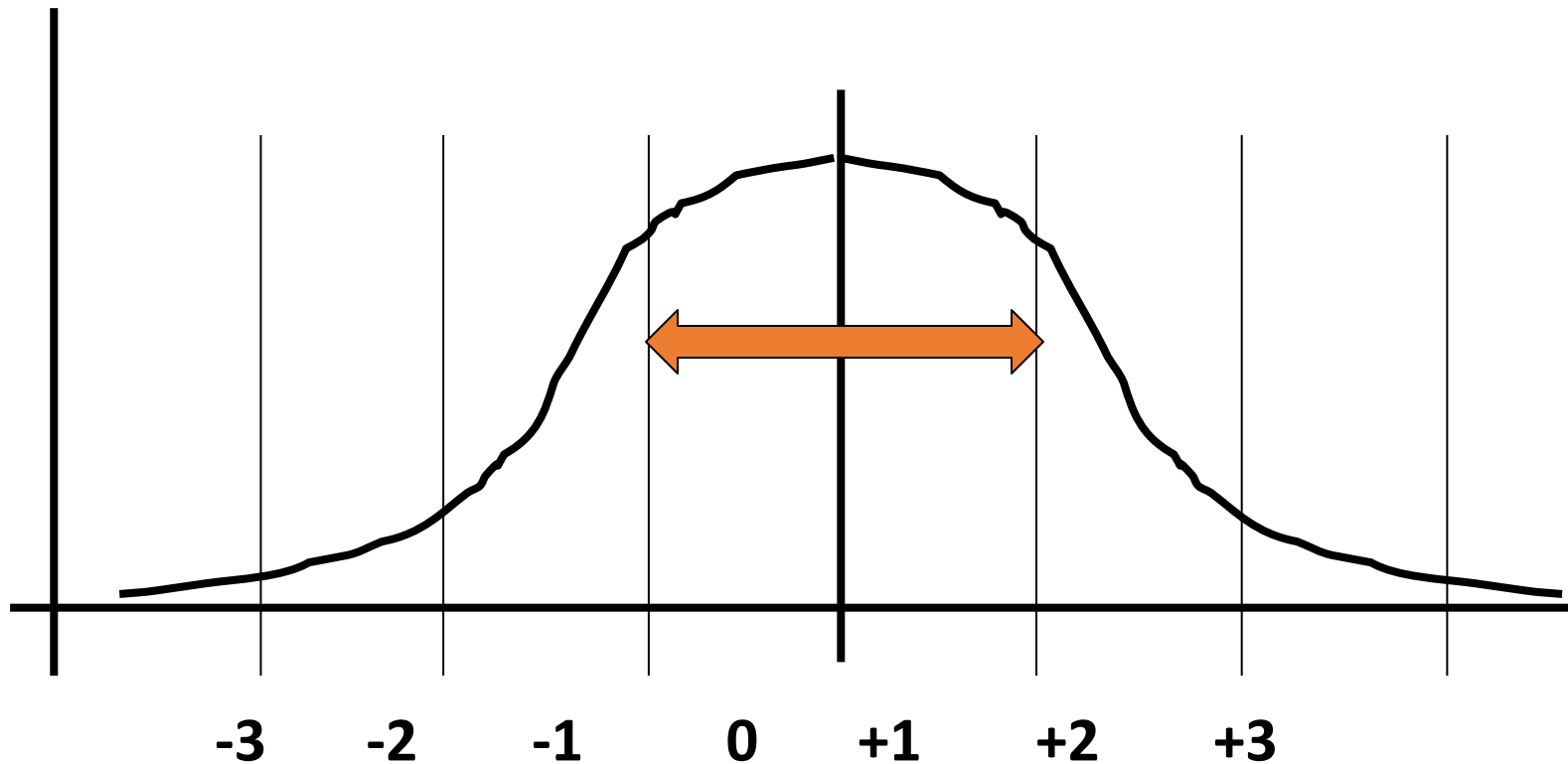
For $X = \$2200$, $Z = (2200 - 2000) / 200 = 1.0$

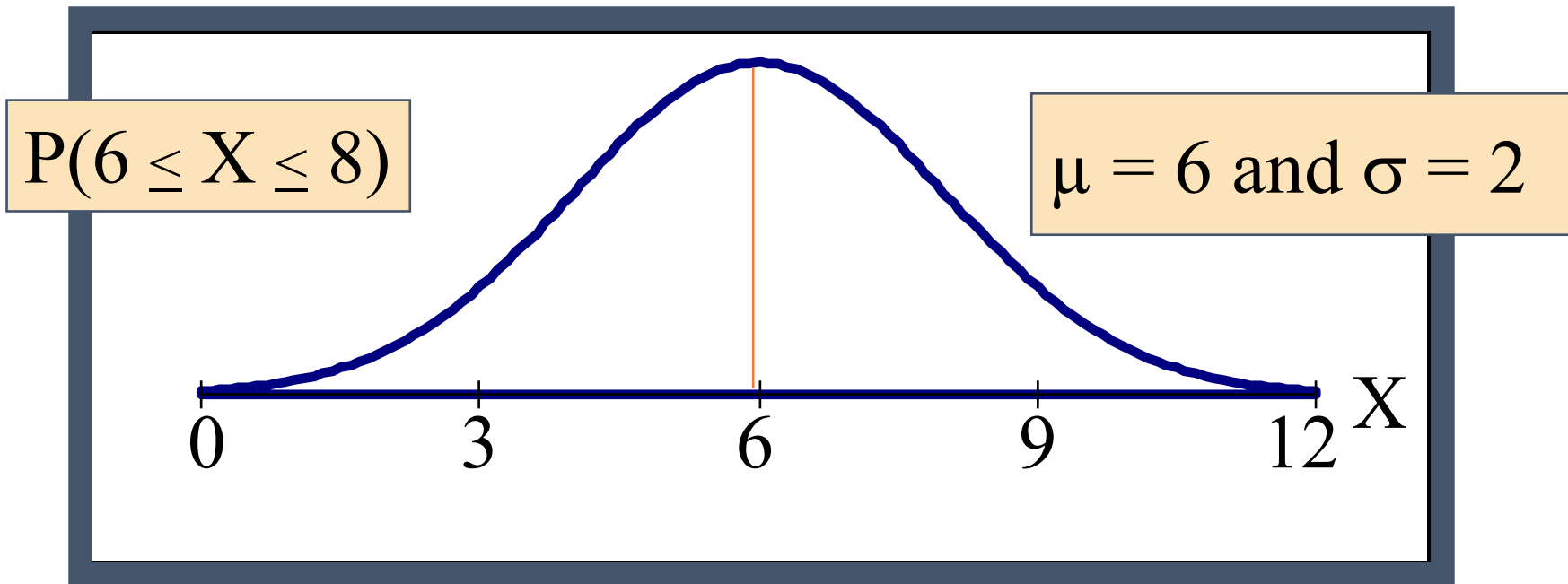
For $X = \$1800$, $Z = (1800 - 2000) / 200 = -1.0$

What does that mean?????

A Z value of 1 indicates that the value of \$2200 is 1 standard deviation **above** the mean of \$2000,

A Z value of \$1800 is 1.0 standard deviation **below** the mean of \$2000.





Probability = area under the density curve

$P(a \leq X \leq b) =$ area under the density curve
between a and b

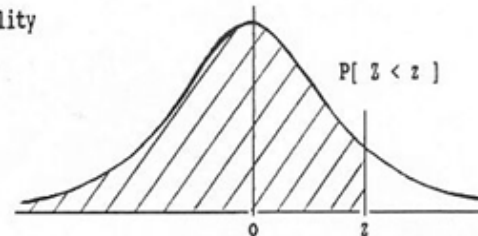
Looking up probabilities in the standard normal table

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability up to the standardised normal value z i.e.

$$P[Z < z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right) dz$$



What is the area to the left of $Z=1.51$ in a standard normal curve?

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5159	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7854
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8804	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767

$Z=1.51$

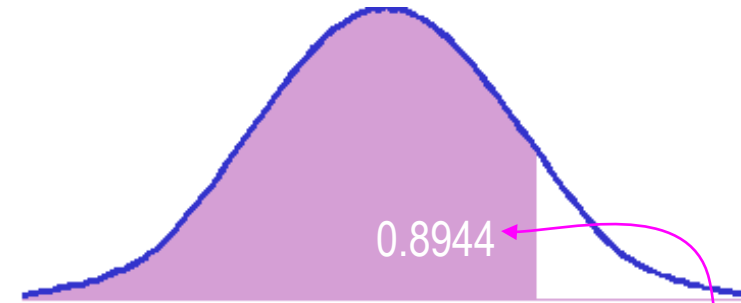
$Z=1.51$

Area is 93.45%

- Example $-X \sim N(60, 64)$

$$P(X < 70) = P\left(\frac{X - 60}{8} < \frac{70 - 60}{8}\right)$$

$$= P(Z < 1.25)$$



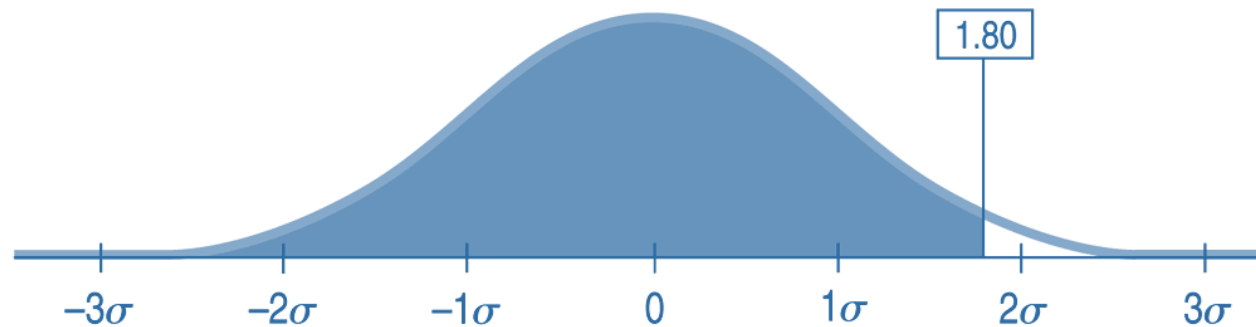
$$P(Z < 1.25) = 0.8944$$

In this example $z_0 = 1.25$

<i>z</i>	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
⋮			⋮			⋮				⋮
1	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
⋮			⋮			⋮				⋮
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
⋮			⋮			⋮				⋮

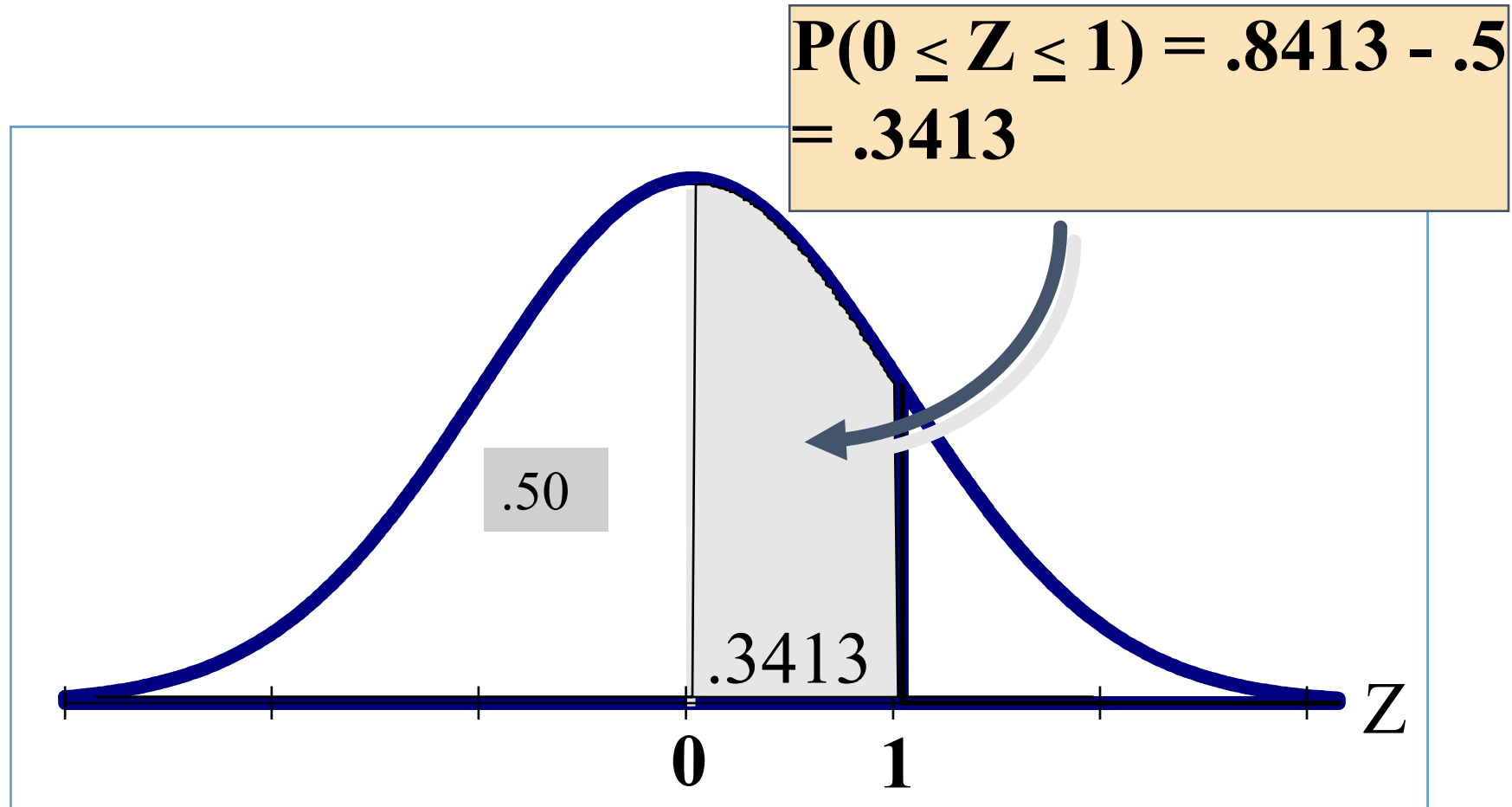
Finding Normal Percentiles by Hand (cont.)

- Table Z is the standard Normal table. We have to convert our data to z-scores before using the table.
- The figure shows us how to find the area to the left when we have a z-score of 1.80:

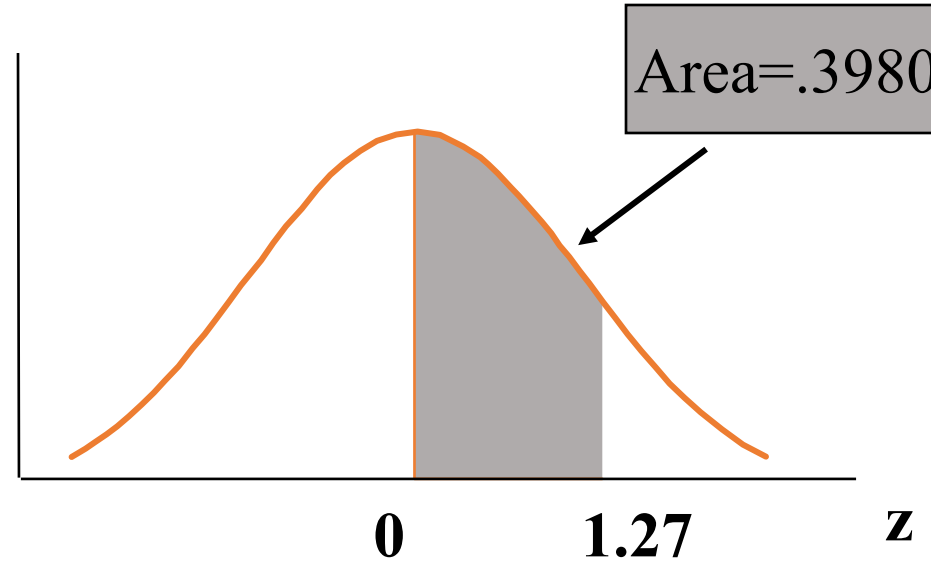


z	.00	.01
⋮	↓	⋮
1.7	.9554	.9564
1.8 →	.9641	.9649
1.9	.9713	.9719
⋮	⋮	⋮

Areas Under the Z Curve: Using the Table



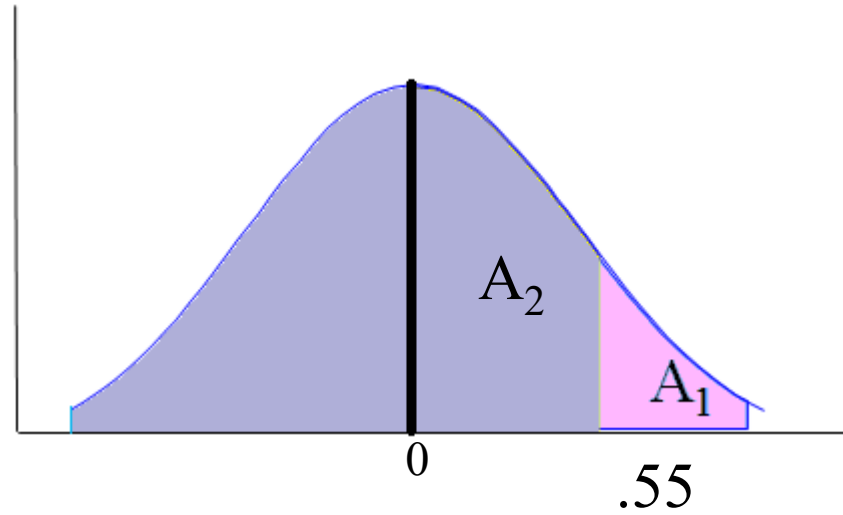
Examples



- $P(0 \leq z \leq 1.27) =$

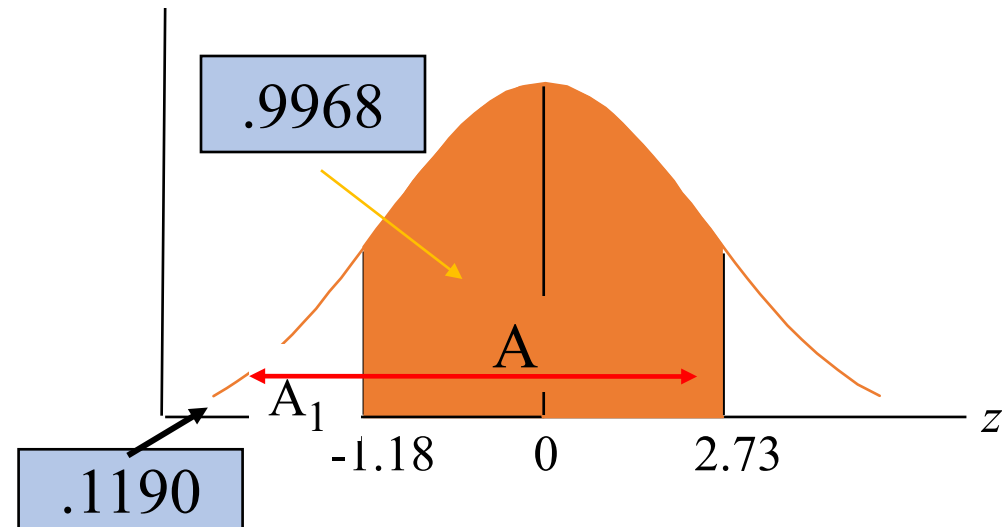
$$.8980 - .5 = .3980$$

Example

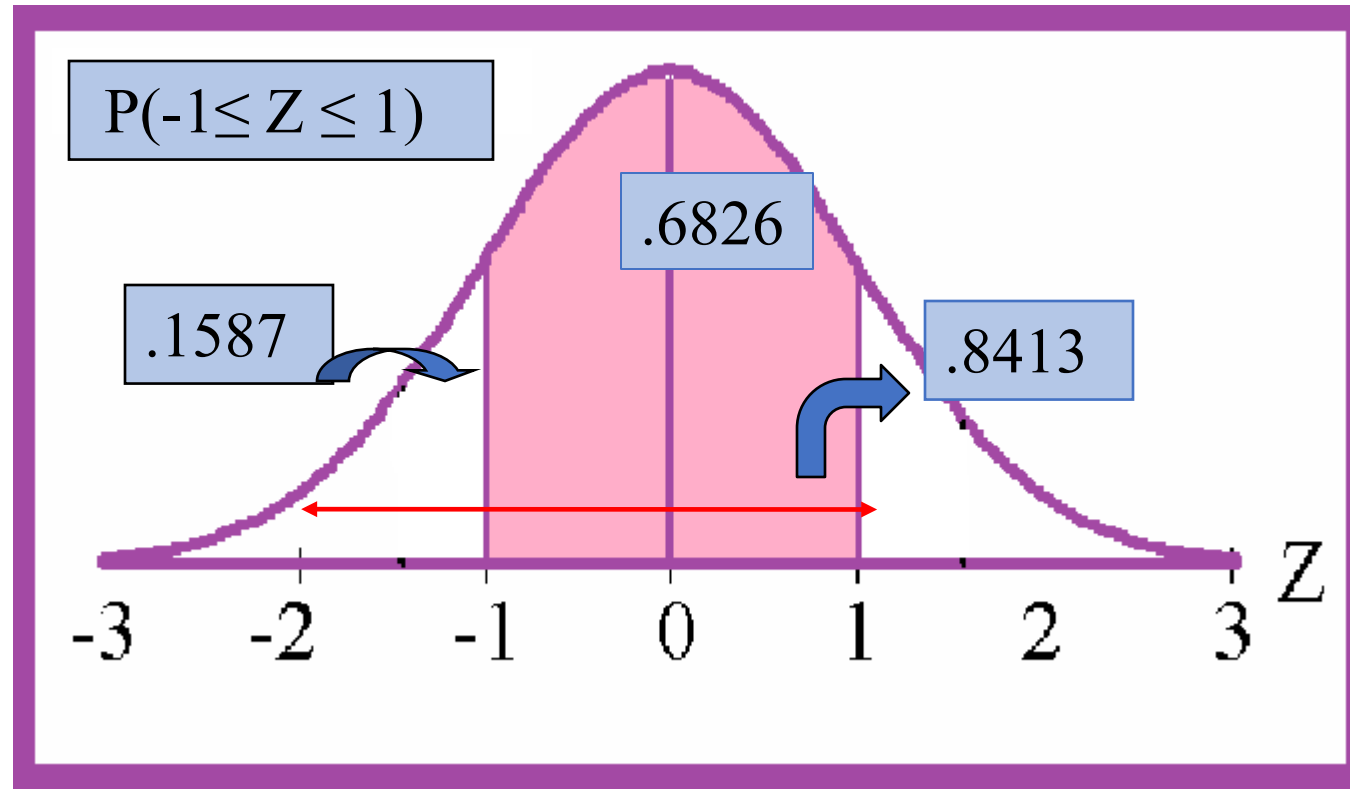


$$\begin{aligned} P(Z \geq .55) &= A_1 \\ &= 1 - A_2 \\ &= 1 - 0.7088 \\ &= 0.2912 \end{aligned}$$

Examples.

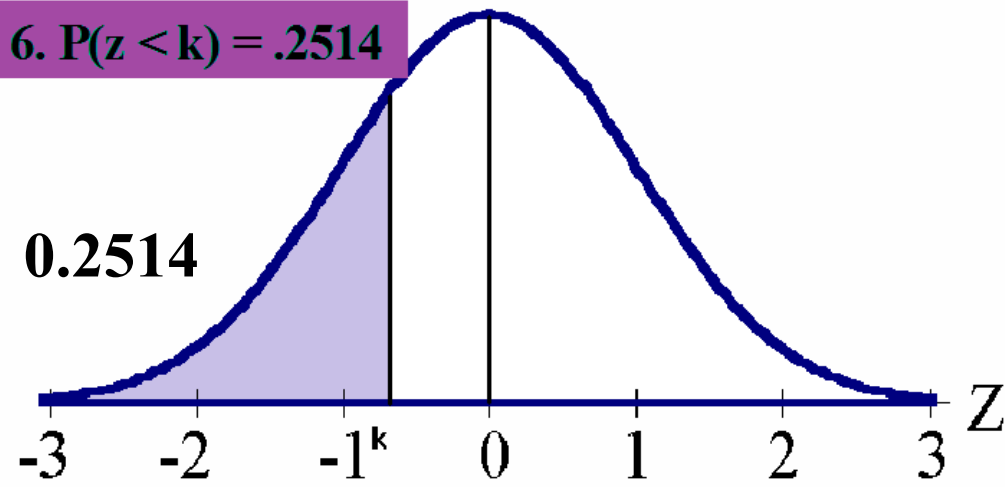


- $P(-1.18 \leq z \leq 2.73) = A - A_1$
- $= .9968 - .1190$
- $= .8778$



$$P(-1 \leq Z \leq 1) = .8413 - .1587 = .6826$$

6. $P(z < k) = .2514$



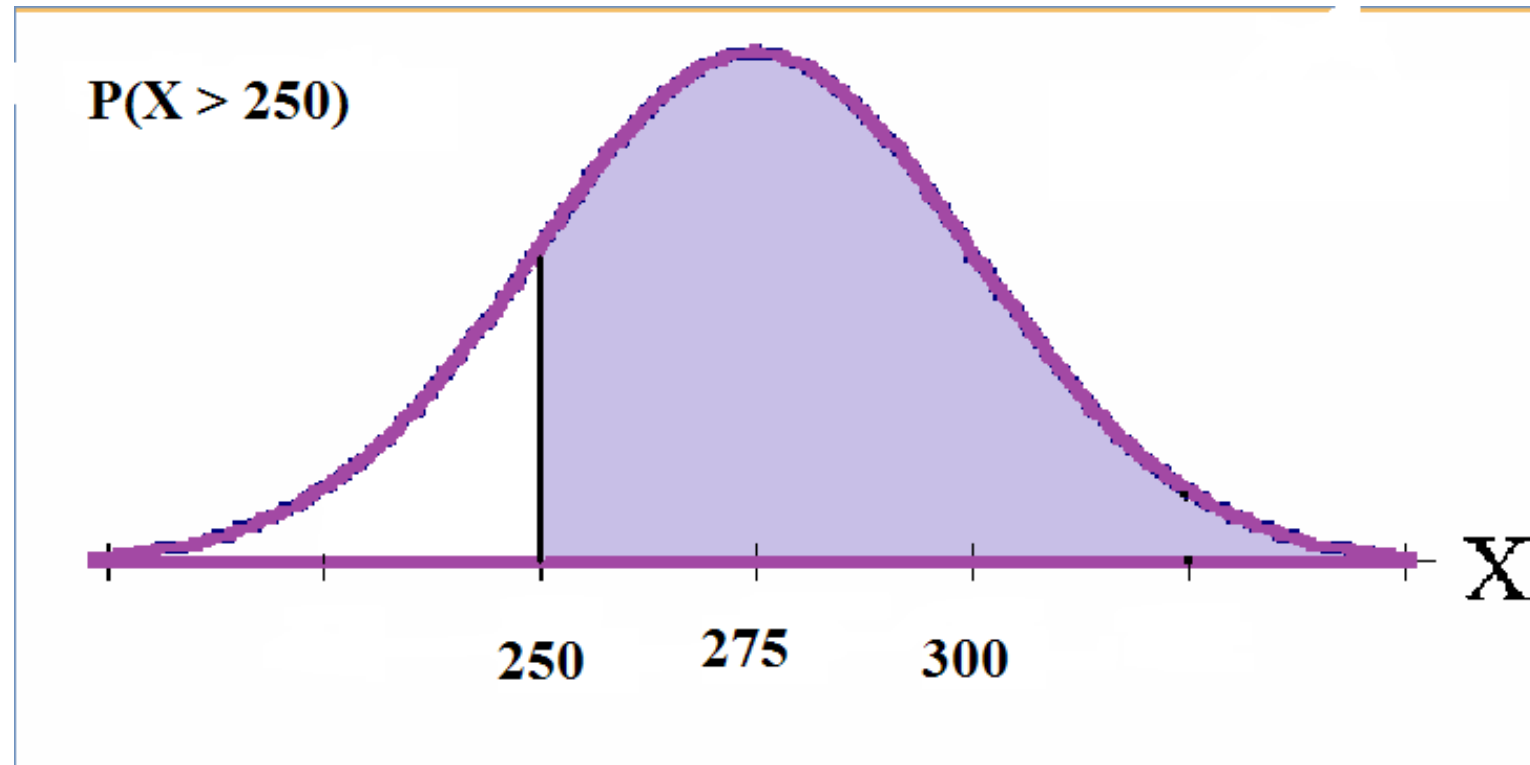
z	.00	.01	.02	.03	.04	.05	.06	.07	.08
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810

Is k positive or negative?

magnitude of probability

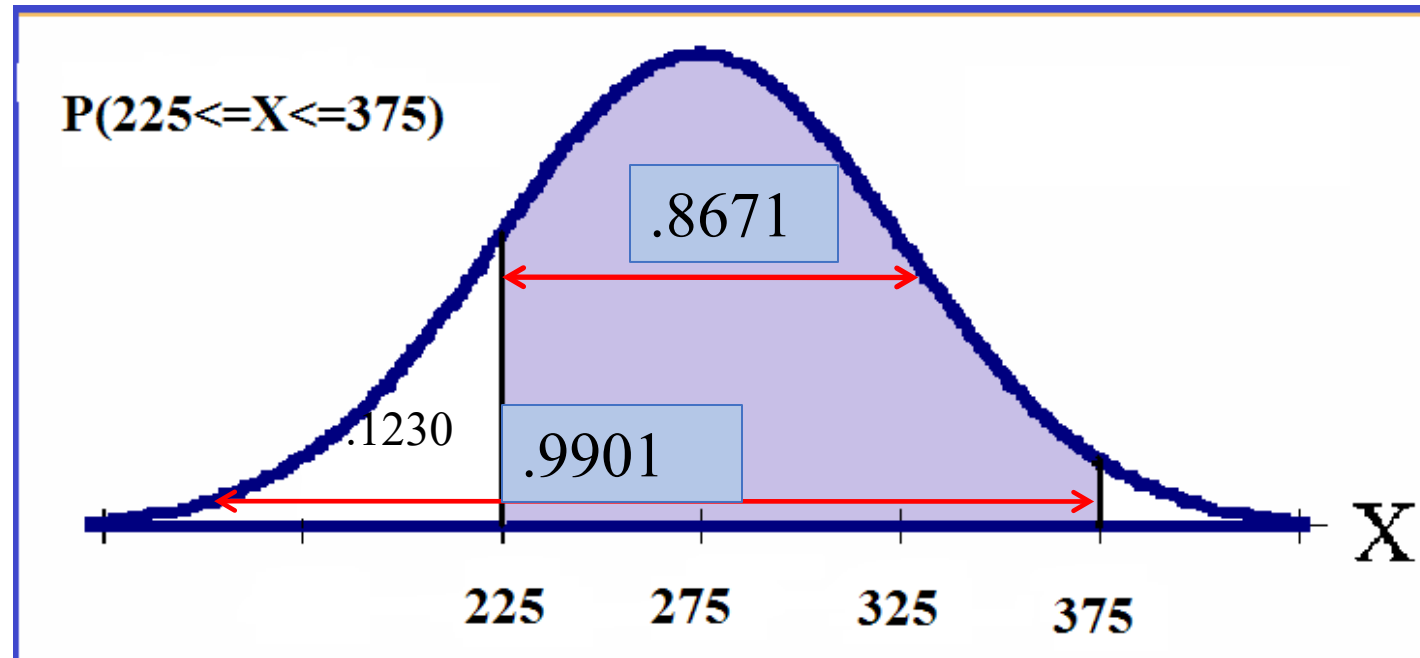
Look up 0.2514 in the z table; corresponding entry is -0.67

Examples $X \sim N(275, 1849)$



$$\begin{aligned} P(Z > \frac{-25}{43}) &= P(Z > -0.58) = 1 - 0.2810 \\ &= 0.7190 \end{aligned}$$

Examples



$$\begin{aligned} \diamond P(225 \leq x \leq 375) \\ &= P\left(\frac{225 - 275}{43} \leq \frac{x - 275}{43} \leq \frac{375 - 275}{43}\right) \\ &= P(-1.16 \leq z \leq 2.33) = 0.9901 - 0.1230 = 0.8671 \end{aligned}$$

$$\begin{aligned} \diamond P(x \leq 225 \text{ or } x \geq 375) \\ &= P(x \leq -1.16 \text{ or } x \geq 2.33) \\ &= 1 - P(-1.16 \leq z \leq 2.33) = 1 - 0.8671 = 0.1329 \end{aligned}$$

Exercises

- Assuming the normal heart rate (H.R) in normal healthy individuals is normally distributed with Mean = 70 and Standard Deviation =10 beats/min

Exercise

Then:

What area under the curve is under 80 beats/min?

And what area under the curve is above 80 beats/min?

- Now we know, $Z = (X - M) / SD$

$Z = ?$ $X = 80$, $M = 70$, $SD = 10$.

So we have to find the value of Z . For this we need to draw the figure.....and find the area which corresponds to Z .

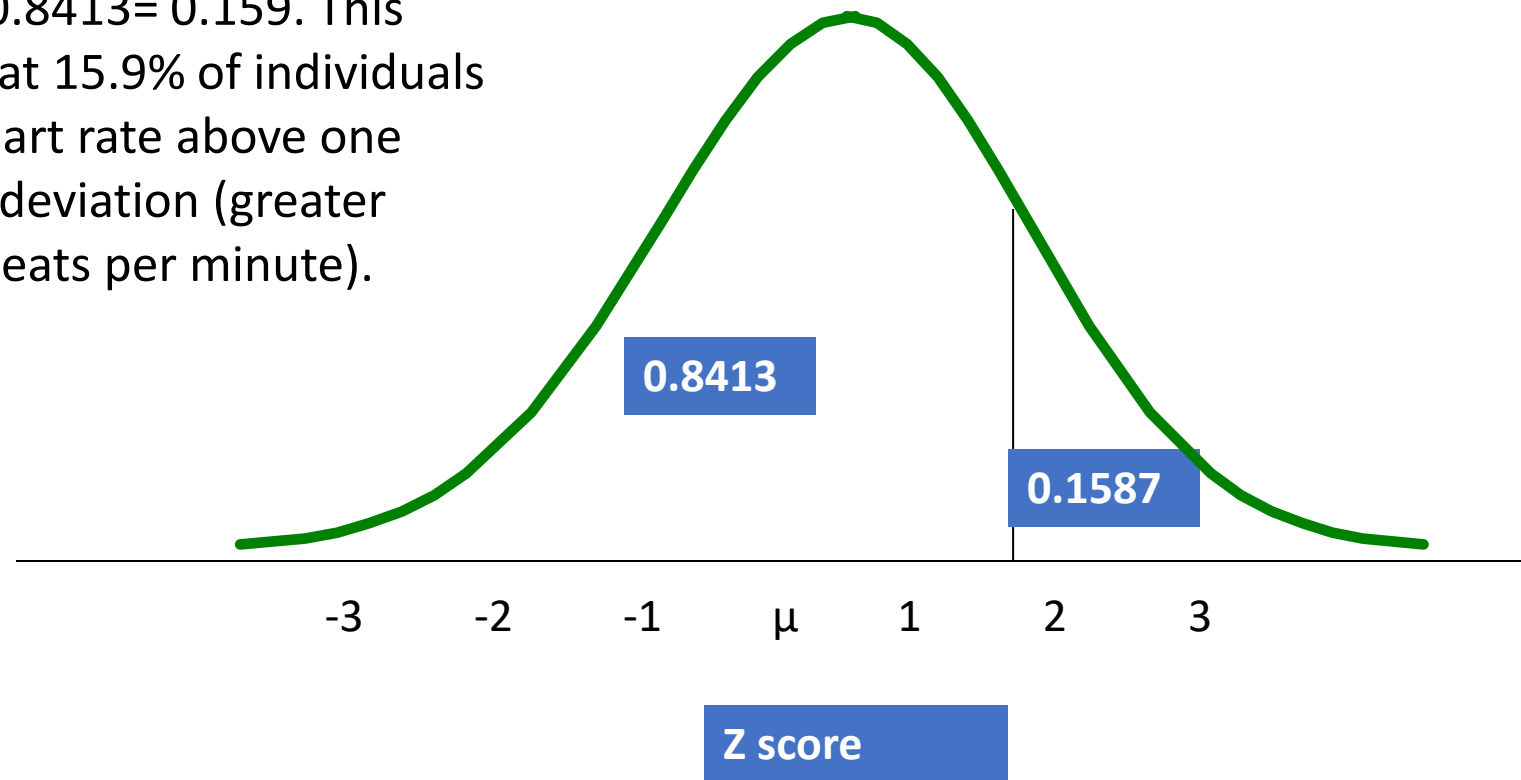
$Z = 1$ then one SD above the mean.

The value of z from the table for $z=1.00$ is 0.8413.

So 84% have heart rate of 80 beats or less per minute.

Then $1 - 0.8413 = 0.1587$. This means that 15.9% of individuals have a heart rate above one standard deviation (greater than 80 beats per minute).

Exercise



Then:

2) What area of the curve is below 90 beats/min?

And what area of the curve is above 90 beats/min?

Solution:

Find Z score then for $x=90$

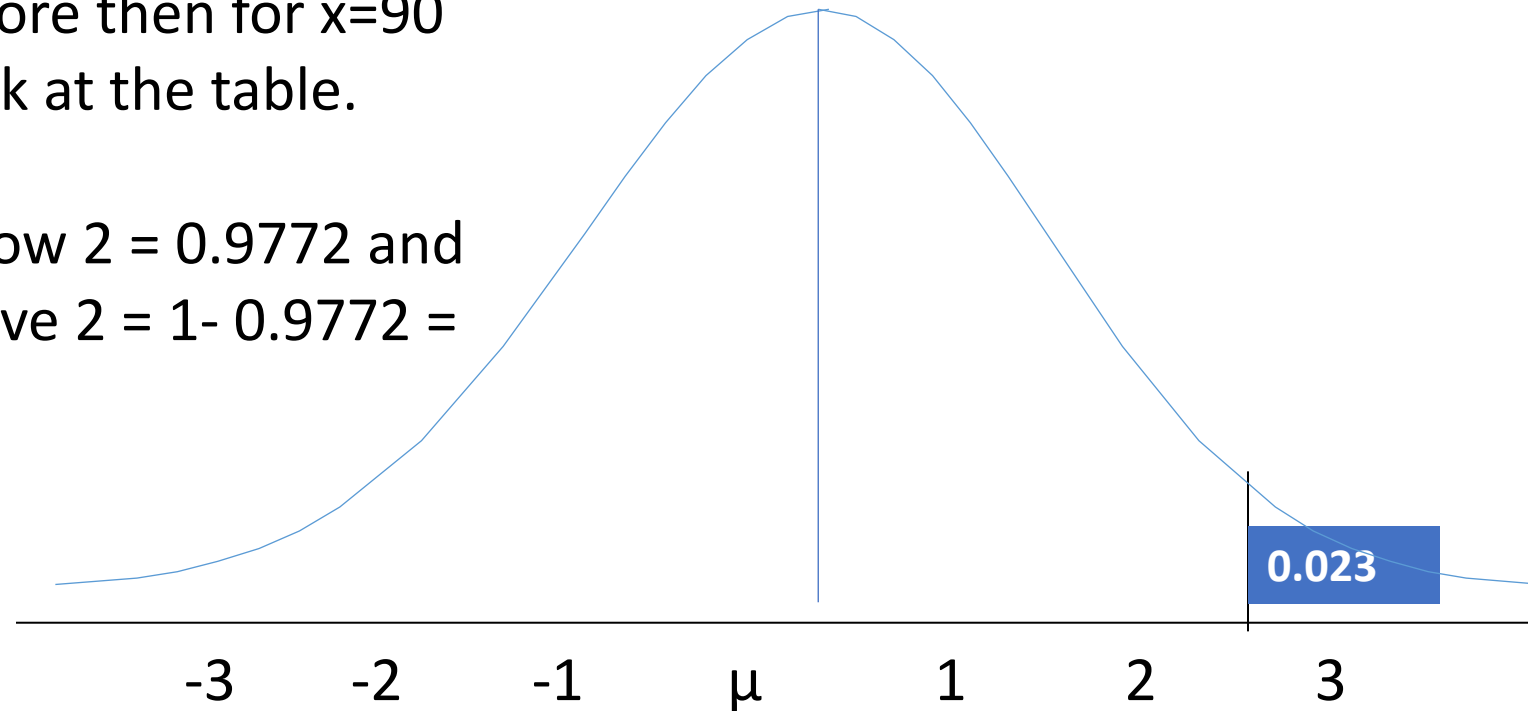
Then look at the table.

$Z = 2$

Area below 2 = 0.9772 and

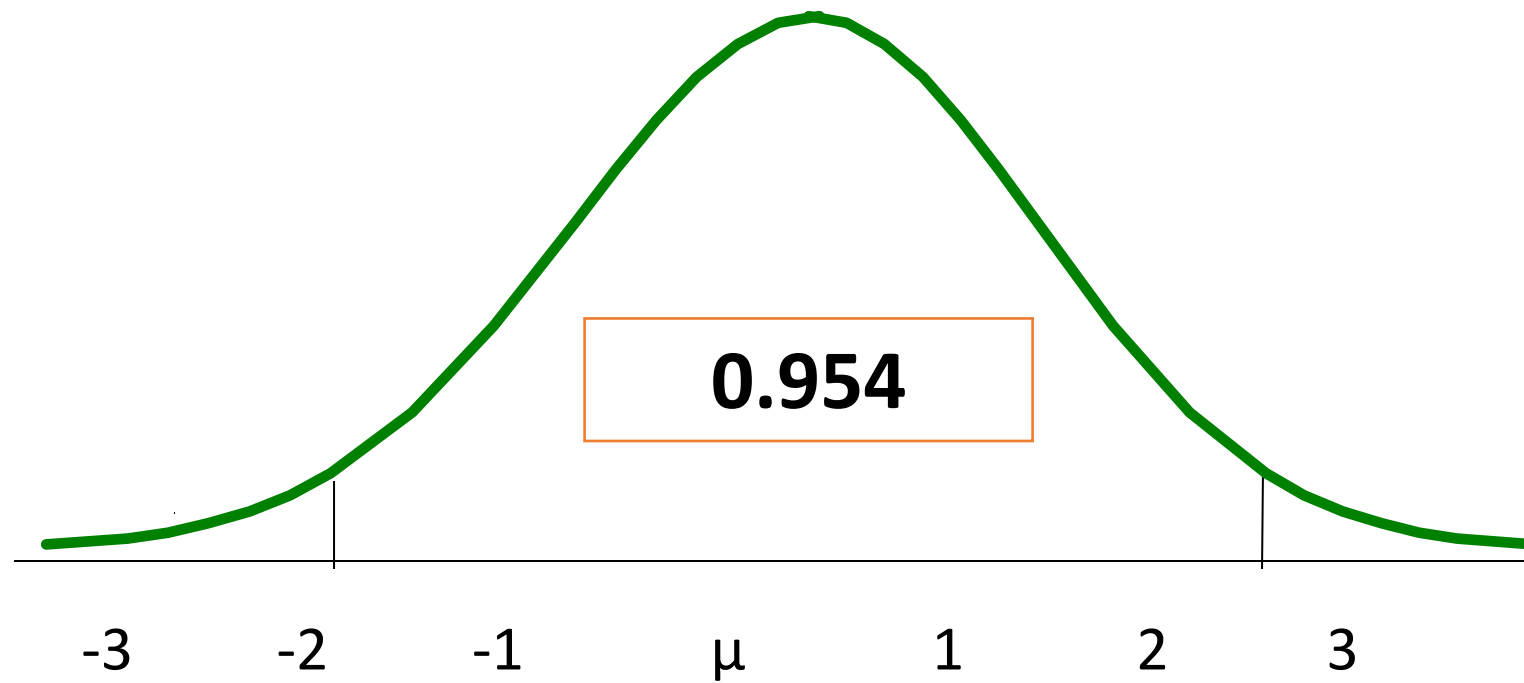
area above 2 = $1 - 0.9772 =$

0.023



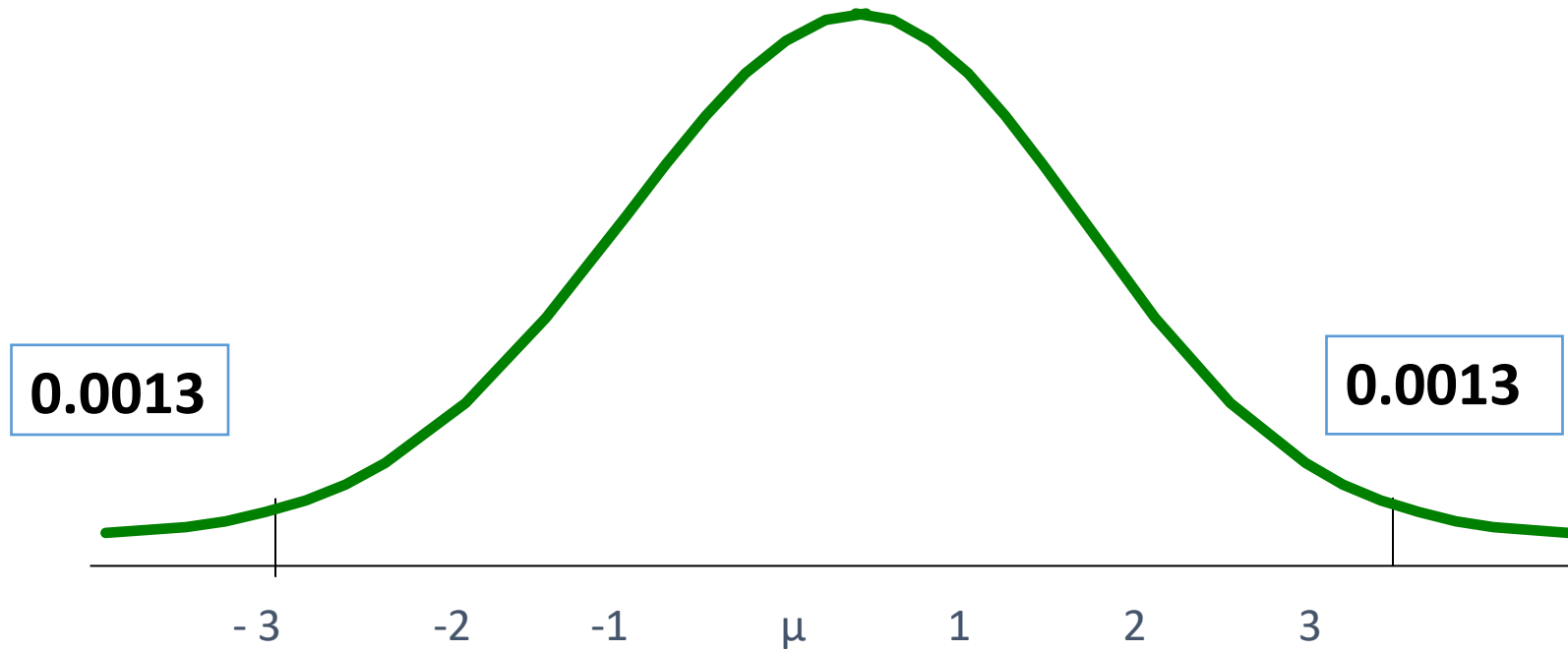
Exercise

- 3) What area of the curve is between 50-90 beats/min?



Exercise

4) What area of the curve is below 40 beats per min or above 100 beats per min? $0.0013 + 0.0013 = 0.0026$

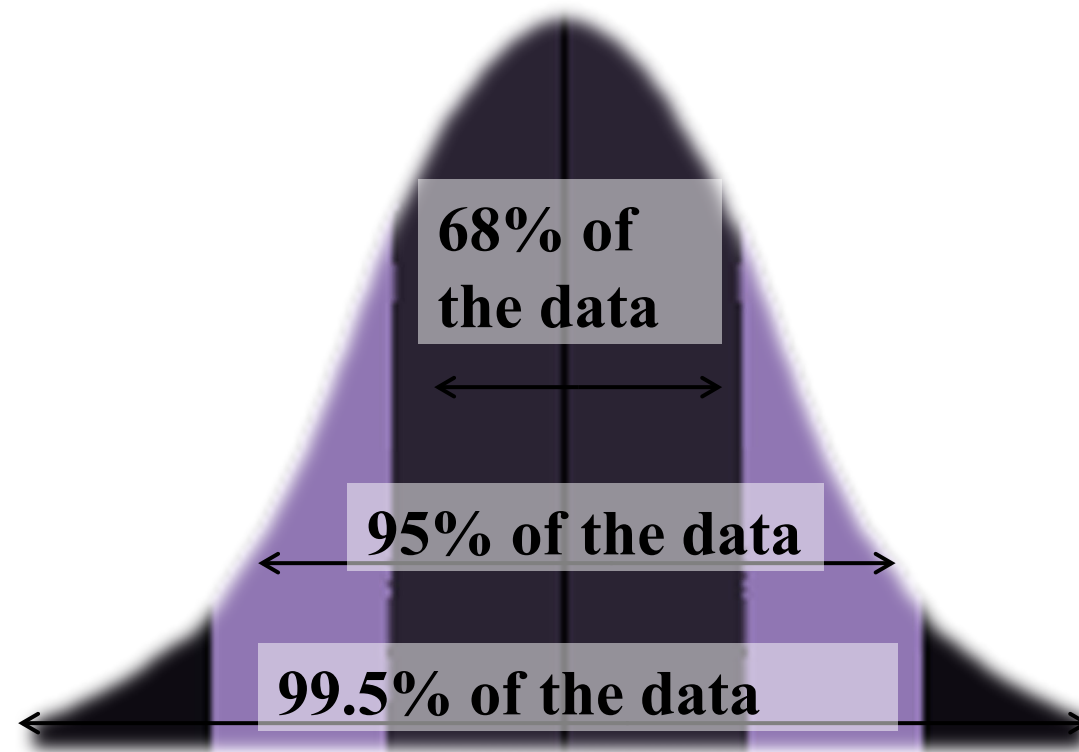


Distinguishing Features

- The mean ± 1 standard deviation covers 68% of the area under the curve
- The mean ± 2 standard deviation covers 95% of the area under the curve
- The mean ± 3 standard deviation covers 99.5% of the area under the curve



68-95-99.5 Rule



Are my data normally distributed?

1. Look at the histogram! Does it appear bell shaped?
2. Compute descriptive summary measures—are mean, median, and mode similar?
3. Do 2/3 of observations lie within 1 std dev of the mean? Do 95% of observations lie within 2 std dev of the mean?
4. Q-Q PLOTS